

Data Cleaning, Transformation & Visualization with Power BI



Project Objective:

This project focuses on **data transformation, cleaning, modeling, and visualization in Power BI** to generate meaningful business insights from sales data. The workflow includes importing multiple datasets, performing ETL operations, creating relationships between tables, developing calculated fields using DAX, and designing an interactive dashboard for decision-making.



Dataset Description:

This dataset contains sales, customer, product, and order-related information used to perform data transformation, data modeling, and business analysis in Power BI. The purpose of the dataset is to analyze business performance, identify sales trends, evaluate profitability, and generate actionable insights through interactive dashboards.

The dataset includes multiple related tables such as:

- **Orders Table** – Contains order details including Order ID, Order Date, Customer ID, Product ID, Quantity, Sales, and Discount.
- **Customers Table** – Stores customer information such as Customer ID, Customer Name, Segment, Region, and other demographic details.
- **Products Table** – Includes product-related details like Product ID, Product Name, Category, and Sub-Category.
- **Sales/Profit Table** – Contains revenue, cost, and profit-related metrics used for financial analysis.
- **Targets Table (if applicable)** – Stores sales targets for performance comparison and KPI tracking.

Key Objectives of the Dataset

- Clean and transform raw business data using ETL processes.
- Build relationships between multiple tables through data modeling.
- Create calculated columns and DAX measures.
- Analyze sales performance and profitability.
- Develop interactive dashboards and reports for business decision-making.

Tools Used

- **Power BI Desktop** – Data Modeling & Dashboard Development
- **Power Query Editor** – Data Cleaning & Transformation
- **DAX (Data Analysis Expressions)** – Calculations and KPI creation



Column Description

Column Name	Description
Order ID	Unique identifier for each order
Order Date	Date when the order was placed
Customer ID	Unique identifier for each customer
Customer Name	Name of the customer
Product ID	Unique identifier for each product
Product Name	Name of the purchased product
Category	Product category (Electronics, Clothing, etc.)
Quantity	Number of units purchased
Unit Price	Price per unit of the product
Discount	Discount applied to the order
Revenue	Total sales amount generated
Cost	Total cost of products sold
Profit	Revenue minus cost
Region	Geographic sales region
Sales Target	Expected sales value



Data Cleaning & Transformation

- Imported raw datasets into Power BI and verified data types.
- Removed duplicate records to maintain data accuracy.
- Handled missing values using replacement rules and transformations.
- Standardized text fields (customer names, categories, IDs).
- Corrected inconsistent formats and renamed columns for clarity.
- Created calculated columns and measures using DAX where required.
- Established relationships between tables for efficient data modeling.
- Applied transformations using Power Query Editor:
 - **Change Data Types**

1	B-2	168	
2	B-2		
3	B-2		
4	B-2		
5	B-25602	168	

	\$ Amount	123
%	Valid	100%
%	Error	0%
%	Empty	0%
		1,275.00
		66.00
		8.00
		80.00
		168.00
		424.00

○ Split/Merge Columns

Merge Columns

Choose how to merge the selected columns.

Separator
--None--

New column name (optional)
Merged

OK Cancel

```
,{"Location", Text.Trim, type text}}
```

A _C State	A _C City	A _C Location
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
Gujarat	Ahmedabad	Ahmedabad,Gujarat
Maharashtra	Pune	Pune,Maharashtra
Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
Rajasthan	Jaipur	Jaipur,Rajasthan
West Bengal	Kolkata	Kolkata,West Bengal
Karnataka	Bangalore	Bangalore,Karnataka
Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
Tamil Nadu	Chennai	Chennai,Tamil Nadu
Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh

○ Conditional Columns

Add a conditional column named Profit Status based on:

- Profit < 0 → *Loss*
- Profit = 0 → *Break-Even*
- Profit > 0 → *Profit*

Transform Add Column View Tools Help

Conditional Column Index Column Duplicate Column Merge Columns Extract Parse From Text Statistics Standard Scientific Rounding Information Date Time Duration From Date & Time

Table: ReorderColumns(#"Changed Type4", {"Order ID", "Amount", "Profit", "Profit Margin", "Quantity", "Category"})

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name: Profit Status

If: Profit is less than 0 Then: Loss

Else If: Profit equals 0 Then: Break - Even

Else If: Profit is greater than 0 Then: Profit

OK Cancel

Transform Add Column View Tools Help

Transpose Reverse Rows Count Rows Data Type: Text Detect Data Type Rename Replace Values Fill Pivot Column Unpivot Columns Move Convert to List Split Column Format Text Column Merge Columns Extract Parse Statistics Standard Scientific Rounding Information Date Time Duration From Date & Time

Table.TransformColumnTypes(#"Reordered Columns1",{{"Profit Status", type text}})

	\$ Amount	\$ Profit	% Profit Margin	Profit Status
	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	1,275.00	-1,148.00	-90.04%	Loss
2	66.00	-12.00	-18.18%	Loss
3	8.00	-2.00	-25.00%	Loss
4	80.00	-56.00	-70.00%	Loss
5	168.00	-111.00	-66.07%	Loss
6	424.00	-272.00	-64.15%	Loss
7	2,617.00	1,151.00	43.98%	Profit
8	561.00	212.00	37.79%	Profit
9	119.00	-5.00	-4.20%	Loss
10	1,355.00	-60.00	-4.43%	Loss

○ Data Validation

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Manage Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Text Use First Row as Headers Replace Values Transform

Queries [4]

Table.NestedJoin(#"List of Orders", {"Order ID"}, #"Order Details", {"Order ID"}, "Order Details", JoinKind.LeftOuter)

	Order ID	Order Date	CustomerName	State	City	Location
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	B-25601	01-Apr-18	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
2	B-25602	01-Apr-18	Pearl	Maharashtra	Pune	Pune, Maharashtra
3	B-25603	03-Apr-18	Jahan	Madhya Pradesh	Bhopal	Bhopal, Madhya Pradesh
4	B-25604	03-Apr-18	Divsha	Rajasthan	Jaipur	Jaipur, Rajasthan
5	B-25605	05-Apr-18	Kasheen	West Bengal	Kolkata	Kolkata, West Bengal
6	B-25606	05-Apr-18	Hazel	Karnataka	Bangalore	Bangalore, Karnataka



Visualization and Insights

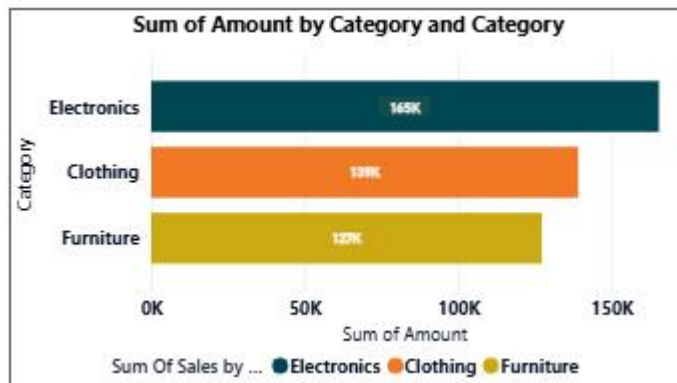
(Attach Screenshot of Visualizations)

Include screenshots of:

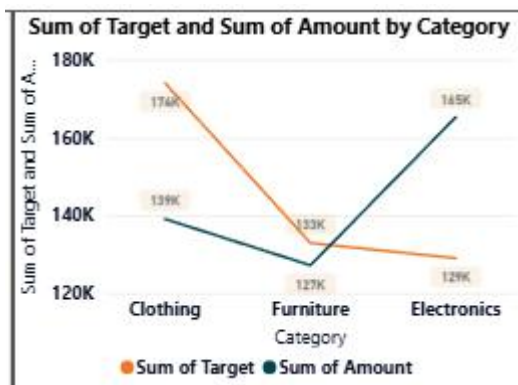
- **KPI Cards**



- **Bar Chart**



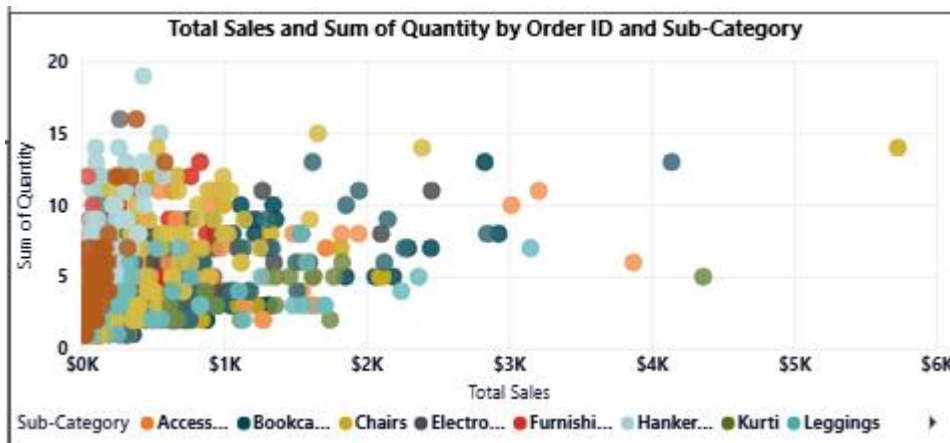
- **Line Chart**



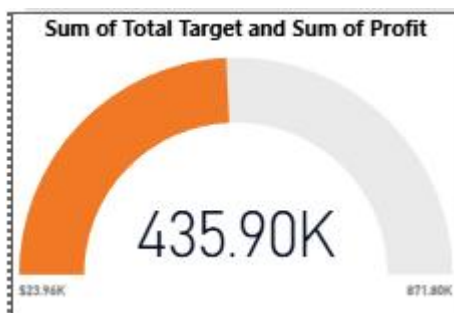
- **Pie/Donut Chart**



- Scatter Chart



- Gauge Chart



- Table

CustomerName	Total Sales
Yohann	\$1,314
Yogesh	\$5,421
Yash	\$791
Yaanvi	\$9,177
Wale	\$130
Vivek	\$114
Vishakha	\$4,836
Vipul	\$16
Vini	\$2,139
Vineet	\$666
Vikash	\$1,350
Total	\$431,502

Key Insights:

- Highest revenue-generating category identified.
- Top-performing products contributed major sales volume.
- Profit varied across categories and regions.
- Seasonal/monthly trends observed in customer purchasing behavior.

Report / Dashboard Screenshot

Insert final dashboard screenshot here.

Dashboard Components:

- KPI Summary
- Sales Performance
- Profit Analysis
- Product Performance
- Interactive Filters



Analysis

1. Descriptive Analysis

Purpose:

Summarize historical data and show overall performance.

- Total Sales achieved: **\$165.267K**
- Total Profit: **435.90K**
- Category-wise sales:
 - Electronics → Highest sales (~156K)
 - Clothing → ~136K
 - Furniture → ~121K
- Customer-wise sales table shows top contributing customers.

Key Insight:

Electronics generated the highest revenue compared to other categories.

2. Diagnostic Analysis

Purpose:

Identify reasons behind performance.

- Compare **Sales vs Target by Category**
 - Electronics exceeded target.
 - Furniture showed lower sales and missed target.
- Scatter chart (**Sales vs Quantity**) shows:
 - Higher quantity orders do not always generate higher sales.
 - Some products generate high sales with fewer quantities.

Key Insight:

Category performance differences and product demand influence overall revenue.

3. Predictive Analysis

Purpose:

Use trends to forecast future outcomes.

- Electronics sales may continue growing if demand remains stable.
- Furniture could continue underperforming unless strategy changes.
- High-performing customers may contribute more future revenue.

Prediction:

If current sales trends continue, Electronics is expected to remain the leading category.

4. Prescriptive Analysis

Purpose:

Recommend actions for improvement.

Recommendations:

- Increase inventory and promotions for Electronics.
- Introduce discounts or campaigns for Furniture.
- Focus customer retention programs on top buyers.
- Optimize product mix based on quantity vs sales relationship.

Action Plan:

- Maintain high-performing categories.
- Improve low-performing segments.
- Use customer insights for targeted marketing.



Conclusion

This project transformed raw data into meaningful business insights using Power BI. Through data cleaning, modeling, visualization, and analytical techniques, trends and performance drivers were identified. The dashboard supports data-driven decision-making by helping stakeholders monitor KPIs, understand customer behavior, and improve business performance